
🌐 Amazon VPC (Virtual Private Cloud) – Explained with Notes

✅ What is Amazon VPC?

Amazon VPC is a **virtual network** that you define in AWS. It allows you to **launch AWS resources** (like EC2, RDS, Lambda) in a **logically isolated environment** with full control over:

- IP addressing
- Subnet creation
- Route control
- Firewall rules
- Internet access
- VPN connectivity

Think of it as your **own private data center** inside AWS.

🧱 Core Components of a VPC

1️⃣ CIDR Block

- Defines the **IP address range** of the VPC (e.g., 10.0.0.0/16)
- Must be **IPv4**, optional IPv6
- Cannot be changed after creation

2️⃣ Subnets

- Divide the VPC's CIDR block into smaller ranges
- Can be **Public** or **Private**
- Subnets are **tied to an AZ** (Availability Zone)

Subnet Type	Description
Public	Has route to Internet Gateway
Private	No direct internet access

3️⃣ Route Table

- Controls how traffic is routed from subnet to other networks
- Each subnet must be associated with **one route table**
- Custom or default route tables

🧭 Example:

Destination Target
10.0.0.0/16 local
0.0.0.0/0 igw-xxxxxxx (for internet access)

4️⃣ Internet Gateway (IGW)

- Allows **outbound and inbound access** to/from the internet
- Must be **attached to the VPC**

- Public subnets use IGW for internet connectivity

5️⃣ NAT Gateway / NAT Instance

- NAT = Network Address Translation
- Allows **private subnet instances** to **initiate** internet traffic (e.g., updates), but **blocks incoming traffic**

Option	Description
NAT Gateway	Managed, scalable, paid
NAT Instance	Self-managed EC2-based NAT

6️⃣ Elastic IP Address

- A **static public IP address** assigned to your AWS account
- Often used for:
  - Bastion hosts
  - NAT Instances
  - High availability services

7️⃣ Security Groups (SG)

- **Virtual firewalls** at the **instance level**

- **Stateful**: return traffic is automatically allowed
- Supports allow rules only (no deny)

🧱 Example:

Inbound: Allow TCP 22 from My IP
Outbound: Allow all

8️⃣ Network ACLs (NACLs)

- Operate at the **subnet level**
- **Stateless**: each rule must be defined for both directions
- Supports both **Allow and Deny** rules

🧱 Example:

Inbound Rule: Allow TCP 80 from 0.0.0.0/0
Outbound Rule: Allow TCP 80 to 0.0.0.0/0

9️⃣ DHCP Option Set

- Provides default network configuration for instances:
  - Domain name
  - DNS servers
- Default AWS-provided or custom (e.g., for Active Directory)

🔟 VPC Peering

- Allows **private connectivity** between **two VPCs**
- Can be same or cross-region/account
- **No transitive routing** (VPC A ↔ VPC B ↔ VPC C doesn't imply A ↔ C)

1️⃣1️⃣ VPC Endpoints

- Connect privately to AWS services without using the internet
- **Types:**
    - **Interface Endpoint** – ENI in your subnet
    - **Gateway Endpoint** – Used for **S3** and **DynamoDB**

1️⃣2️⃣ Bastion Host

- An EC2 instance in a **public subnet**
- Used to **SSH into private EC2** instances
- Requires Elastic IP + tight Security Group

1️⃣3️⃣ VPC Flow Logs

- Capture **IP traffic logs** for:

    - Network interfaces
    - Subnets
    - VPCs
- Stored in **CloudWatch Logs** or **S3**
- Useful for **security analysis and troubleshooting**

1️⃣4️⃣ Transit Gateway

- Central hub to **connect multiple VPCs and on-prem**
- Simplifies hub-and-spoke architecture
- Supports **routing policies, bandwidth control**, etc.

📐 VPC Architecture Types

🧱 Basic 2-Tier Architecture

- Public subnet → Web servers
- Private subnet → Database

🏗️ High Availability VPC

- Public & private subnets across **multiple AZs**
- NAT Gateway in each AZ

- ALB in public subnet

💬 Summary Table

Component	Purpose
VPC	Isolated virtual network
Subnet	IP ranges inside a VPC
IGW	Internet access for public subnet
NAT	Outbound-only internet from private
SG	Stateful firewall (instance-level)
NACL	Stateless firewall (subnet-level)
Peering	VPC-to-VPC connection
Endpoints	Private connection to AWS services
Flow Logs	Capture network logs

🧪 Suggested Hands-On Lab (Optional)

Would you like me to prepare a step-by-step lab where you:

- Create a VPC
- Add public/private subnets
- Launch EC2 instances
- Attach IGW and NAT

- Set up SG/NACL
- Enable VPC flow logs